Optimization Homework

1.1 Choose Your Character

① Robot final pose

$q \in \mathbb{R}^7$ joint angles
$p(q)$ end effector position
p_{goal} required position
q_{ref} preferred joint posture

$d_j(q)$ signed distance to obstacle (+ve outside obstacle)

Formulation: $\min_q \frac{1}{2} \|q - q_{ref}\|_W^2$

subject to $p(q) = p_{goal}$

$q_{min} \leq q \leq q_{max}$, $d_j(q) \geq 0$ for every j

where $W \succ 0$ weights posture differences

Optimization Algorithm: SQP with BFGS Hessian approx.

→ equality constraints enforce reach
→ other constraints - joint limits & collision avoidance
→ SQP handles smooth non-linear constraints
→ BFGS approximates Hessian with 1st derivatives

(2) 30 Power plants + Market model

Algorithm: ADMM Alternating Direction Method of Multipliers. (Alternatively: Gradient Descent)

Linear predictor $\theta \in \mathbb{R}^{1000}$

Plant i has feature matrix X_i and demand vector y_i

Define $\ell_i(\theta) = \frac{1}{2} \| X_i \theta - y_i \|_2^2$ and we solve

$$\min_{\theta} \sum_{i=1}^{30} \ell_i(\theta) + \lambda \|\theta\|_1 \quad , \lambda > 0$$

We use squared loss to fit demand with ℓ_1 penalty

local copies θ_i and common vector z, $\theta_i = z$

For penalty $\rho > 0$, scaled consensus ADMM

$$\theta_i^{k+1} = \arg\min_{\theta_i} \left\{ \ell_i(\theta_i) + \frac{\rho}{2} \left\| \theta_i - z^k + u_i^k \right\|_2^2 \right\}$$

$$z^{k+1} = S_{\lambda/30\rho} \left(\frac{1}{30} \sum_{i=1}^{30} (\theta_i^{k+1} + u_i^k) \right),$$

$$u_i^{k+1} = u_i^k + \theta_i^{k+1} - z^{k+1}$$

Here, $S_T(v)_j = \text{sign}(v_j) \max(|v_j| - T, 0)$

Plants solve local problems in parallel & communicate parameter-sized vectors per iteration (w/o raw observations)

(3) Sensor + Noisy Readings

Optimization Algo: Null space Reduction + cached QR factorization

Let $r \in \mathbb{R}^{1000}$ be a nonzero reading vector
Normalize it as $y = r/\|r\|_2$

990 independent calibrations : $Cx = b$ $C \in \mathbb{R}^{990 \times 1000}$ has full row rank

$$\min_x \frac{1}{2}\|x - y\|_2^2 \text{ subject } Cx = b$$

Use QR to find minimum norm solution x_p of $Cx = b$ and an orthogonal basis $Z \in \mathbb{R}^{1000 \times 10}$ of $\ker C$

Write $x = x_p + Zz$ then $Z^T x_p = 0$

Objective becomes $\frac{1}{2}\|z - Z^T(y - x_p)\|_2^2 +$ constant giving,

$$z^* = Z^T(y - x_p), \quad x^* = x_p + ZZ^T(y - x_p)$$

Cache the factorization

(4) Plant control - 6 settings

Algorithm: Sparse Primal-Dual Interior Point method.

Let $u_k \in \mathbb{R}^6$: settings and $x_k \in \mathbb{R}^{20}$ process state @ interval k

Stack the local process equations F_k

$$\min_{\{x_k, u_k\}} \sum_{k=0}^{1999} L_k(x_k, u_k) + L_T(x_{2000})$$

subject to $x_0 = x_{init}$, $x_{k+1} = F_k(x_k, u_k)$, $k = 0, 1, \ldots 1999$

Retain states & controls to preserve temporal & neighboring process sparsity.

Use sparse primal dual interior point solver with sparse Jacobians & Lagrangian Hessian

(5) Battery conductivity

Algorithm: Augmented Lagrangian with L-BFGS-B

Let $K \in \mathbb{R}^{100000}$ cell conductivities

$F_i(K)$: predicted data y_i : measurements

E contains adjacent cell pairs
H 5 highly accurate samples.

$$\min_{K_{min} \le K \le K_{max}} f(K) := \frac{1}{2} \sum_{i \in H} \left(F_i(K) - y_i\right)^2 + \frac{\alpha}{2} \sum_{(j,\ell) \in E} (K_j - K_\ell)^2$$

subject to $h_i(K) := F_i(K) - y_i = 0, \quad i \in H$

with $\alpha > 0$. Neighbor penalty encourages similar adjacent conductivities

→ 5 trusted samples ⇒ equalities
→ Minimize augmented Lagrangian $f(K) + v^T h(K) + \beta/2 \, \|h(K)\|_2^2$ within bounds using L-BFGS-B
→ Update $v \leftarrow v + \beta h(K)$ and $\uparrow \beta$ when needed

1st derivatives + limited memory, avoiding Hessian.

(1.02) Can't Go Over It

For fixed $\delta > 0$, the quadratic continuation is

$$q_\delta(s) = \begin{cases} -\log s & s \geq \delta \\ -\log\delta - \frac{s-\delta}{\delta} + \frac{(s-\delta)^2}{2\delta^2} & s < \delta \end{cases}$$

① show that $q_\delta \in C^2(\mathbb{R})$

Differentiate each branch

$$q'_\delta(s) = \begin{cases} -1/s & s \geq \delta \\ (s-2\delta)/\delta^2 & s < \delta \end{cases} \qquad q''_\delta(s) = \begin{cases} 1/s^2 & s \geq \delta \\ 1/\delta^2 & s < \delta \end{cases}$$

At the join, both branches give same value & first 2derivative ($s = \delta$)

$$q_\delta(\delta) = -\log\delta \quad q'_\delta(\delta) = -1/\delta \quad q''_\delta(\delta) = 1/\delta^2$$

Each branch is smooth on interior & at $s = \delta$ the matching conditions establish $q_\delta \in C^2\mathbb{R}$

(2) Tight Lipschitz constants for q_δ' and q_δ''

First derivative, Since $0 < q_\delta''(s) \leq \delta^{-2}$ everywhere,

we use MVT $|q_\delta'(s) - q_\delta'(t)| \leq 1/\delta^2 |s-t|$

Tight on $s < \delta$, q_δ' is linear with slope $1/\delta^2$ Thus,

$$\boxed{\text{Lip}(q_\delta') = 1/\delta^2}$$

Second derivative, Below δ, slope is 0, Above δ

$$\frac{d}{ds} q_\delta''(s) = -2/\delta^3 \quad , \quad \left|\frac{1}{ds} q_\delta''(s)\right| \leq 2/\delta^3$$

Continuity at δ, lets us split any interval crossing the join so the same bound holds. Tight as

$$\lim_{\epsilon \to 0} \frac{|q_\delta''(\delta+\epsilon) - q_\delta''(\delta)|}{\epsilon} = 2/\delta^3$$

$\therefore$ Lipschitz with $\boxed{\text{Lip}(q_\delta'') = 2/\delta^3}$

③ Behavior as $\delta \to 0$

For each fixed $s > 0$, eventually $\delta < s$ so

$$\left(q_\delta(s), q_\delta'(s), q_\delta''(s)\right) = \left(-\log s, -1/s, 1/s^2\right)$$

At $s = 0$, this is $-\log \delta + 3/2$, $-2/\delta$ and $1/\delta^2$

For fixed $s < 0$,

$$q_\delta(s) = -\log \delta + \frac{3}{2} - \frac{2s}{\delta} + \frac{s^2}{2\delta^2}$$

$$q_\delta'(s) = \frac{s - 2\delta}{\delta^2}$$

$$q_\delta''(s) = \frac{1}{\delta^2}$$

Thus for every fixed $s \leq 0$, $q_\delta(s) \to +\infty$

$q_\delta'(s) \to -\infty$

$q_\delta''(s) \to +\infty$

(1.3) Pareto Suboptimal

$$\min_{x \in [0,1]} \left(f_1(x), f_2(x)\right) \quad f_1(x) = x \quad f_2(x) = 1 - x^2$$

① True Pareto front : Pareto optimal if no other feasible point improves one objective w/o worsening another

For any $0 \le x \le y \le 1$,

$$f_1(x) < f_1(y) \qquad f_2(x) > f_2(y)$$

Smaller point better for f_1
Larger ⟶ f_2

} Every feasible decision is Pareto optimal

$$P_{decision} = [0,1], \quad P_{front} = \{(x, 1-x^2) : 0 \le x \le 1\}$$

Front: Entire decreasing curved arc from (0,1) to (1,0)

② Weighted sum scalarization :

Standard non negative weighted sum with $\lambda \in [0,1]$

$$\min_{0 \le x \le 1} g_\lambda(x), \quad g_\lambda(x) = \lambda x + (1-\lambda)(1-x^2)$$

Curvature satisfies $g_\lambda''(x) = -2(1-\lambda) \le 0$

For $\lambda < 1$, strictly concave. minimum at endpoint

For $\lambda = 1$, function : x and minimized at $x = 0$

Endpoint values: $g_\lambda(0) = 1 - \lambda$ & $g_\lambda(1) = \lambda$

$\therefore$

$$\arg\min_{x \in [0,1]} g_\lambda(x) = \begin{cases} \{1\} & 0 \leq \lambda < 1/2 \\ \{0,1\} & \lambda = 1/2 \\ \{0\} & \frac{1}{2} < \lambda \leq 1 \end{cases}$$

At equal weights internal points are not minimizers

because $g_{1/2}(x) = \frac{1}{2} + \frac{1}{2} x(1-x) \geq 1/2$ for $0 < x < 1$

Weighted sum recovers two endpoints
But every point on curve is pareto optimal.

concave shape $\longrightarrow$ prevents interior pt. selection.

sol 6.1

AI Usage Acknowledgement : I used ChatGPT to understand problems, theory & help with solutions where I was stuck.